

PROJECT REQUIREMENTS

Project Name

AI Learning Environment (working name)

Functional Requirements

Fall 2026 (ENSE 400)

- **FR-1 Course folders:** the student can add local course folders, and the app lists the PDFs in each one by course. Access can be removed at any time, and original files are never changed.
- **FR-2 PDF reader:** the student can open a PDF in a built-in reader with page navigation, zoom and text search.
- **FR-3 Passage selection:** the student can select text on a page, or a whole page, as the context for a question.
- **FR-4 Ask by text:** the student can type a question about the selection or ask for a plain explanation of it.
- **FR-5 Page-anchored explanations:** every answer shows the file name and page number it came from, and clicking the reference opens that page.
- **FR-6 Model picker:** the student chooses the model for explanations from a list sorted from least to most expensive, showing each model's price tier. A free model is the default.
- **FR-7 Flashcards:** the app generates at least 5 flashcards from a selection; the student can edit, delete and save them to the course's deck.
- **FR-8 Quizzes:** the app generates a quiz of at least 5 multiple-choice questions from a selection, marks answers instantly and shows the page reference for each question.
- **FR-9 History:** saved explanations, flashcards and quiz results are kept per course and can be reopened later.
- **FR-10 Privacy notice:** before first use, the app explains what text is sent to which AI provider and whether that provider may use it to improve its models.
- **FR-11 Model errors:** if a model is unavailable or rate-limited, the app says so and offers another free model.
- **FR-12 Scanned pages:** if a page has no text layer, the app tells the student that text cannot be selected on that page.

Winter 2027 (ENSE 477)

- **FR-13 Accounts:** the student can sign up and sign in with an email address.
- **FR-14 Pro subscription:** the student can subscribe to, manage and cancel Pro, which unlocks the paid models in the model picker.
- **FR-15 Usage limits:** Pro users see their monthly usage; when the limit is reached, explanations fall back to the free models.
- **FR-16 Spaced review:** the student enters exam dates per course, and the app schedules flashcard review before each exam with a daily review queue.
- **FR-17 Voice input:** the student can ask questions by speaking.
- **FR-18 Text recognition:** scanned pages are converted to text (OCR) so they can be selected and explained.
- **FR-19 iPad version:** the student can read, ask questions and review flashcards on an iPad.

Technical/Performance Requirements

Technical requirements

- **TR-1 Platform:** desktop app built with Electron and React, running on Windows 10/11 and macOS 13 or later. The iPad approach (for example React Native) is chosen at the start of ENSE 477.
- **TR-2 PDF handling:** PDF.js renders pages and extracts the text layer, which provides the page numbers used in references.
- **TR-3 Local storage:** a SQLite database on the device stores the course list, explanations, flashcards and quiz results. Course files are read in place and never copied.
- **TR-4 AI layer:** one model-independent interface, with each provider added as an adapter. Free: Gemini API free tier, Groq, OpenRouter free models. Paid: Anthropic Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5, with others

added later. Adding a model needs no change to the reader or practice features.

- **TR-5 Key security:** no API keys are stored in the desktop app. Requests go through a small server proxy that holds the keys and enforces rate limits, hosted on a free tier.
- **TR-6 Data sent:** only the selected text, its page number, the file name and the question are sent to a model. Whole files are never uploaded, and the server stores only usage counts (plus account and subscription data from ENSE 477).
- **TR-7 Structured output:** flashcards and quizzes are returned as JSON and checked against a schema before they are shown.
- **TR-8 Accounts and payments (ENSE 477):** a hosted authentication and database service (for example Supabase) and a payment processor (for example Stripe). Card details never reach the team's servers.
- **TR-9 Source control:** all code is in the team's GitHub repository, and every change is merged through a pull request reviewed by the other member.

Performance requirements

- **PR-1 Explanation speed:** the first words appear within 3 seconds (streamed), and the median complete answer takes 10 seconds or less on the free model.
 - **PR-2 Practice speed:** a flashcard set or quiz is generated in 20 seconds or less.
 - **PR-3 Reader speed:** a 100-page PDF opens to its first page within 2 seconds on a typical student laptop with 8 GB of memory.
 - **PR-4 Folder scan:** a course folder with 200 PDFs is listed within 10 seconds.
 - **PR-5 Memory:** the app uses less than 500 MB of memory with a 300-page PDF open.
 - **PR-6 Accuracy:** 100% of explanations carry a page reference, and at least 85% are rated accurate against the source page.
 - **PR-7 Fallback:** if a free provider fails, the request is retried on another free provider without the student having to do anything.
 - **PR-8 Usability:** a System Usability Scale (SUS) score of 70 or higher in usability testing.
-